

Certified Geometry-Aware Control Barrier Functions for Non-Convex Robots: From Planar Configuration-Space Fields to Articulated Manipulators

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Abstract—Geometry-aware control barrier functions (CBFs) that evaluate signed distance fields (SDFs) at closest boundary points need convex level sets, treat non-convex geometry through rotation-invariant bounding circles, and rely on a heuristic enclosure margin. We show that collision avoidance between two non-convex rigid bodies reduces to a condition on their boundaries at first contact, and enforce it on the whole continuous boundary without computing closest points. Each body’s SDF is a B-spline whose level enclosing the true geometry is certified on the whole continuous boundary, and the safe set requires the SDF of one body to stay above its level on the certified level set of the other. Offline, this surface is covered by boxes selected from Bernstein coefficients; online, the barrier condition on each box is enforced through the Bernstein coefficients of a lifted barrier, finitely many constraints affine in the inputs, without optimization. The safe set is not discretized: the boxes and the certified Taylor bounds only restrict the admissible inputs, by an amount of second order in the box size. The construction is the same for planar and spatial, rigid and articulated bodies. For planar pairs, the condition can instead be compiled into one configuration-space field per pair, certified for all continuous orientations by a Bernstein branch-and-bound, which trades conservatism for a single barrier per pair. The guarantees hold in continuous time and exact arithmetic. In simulation, both constructions prevented all collisions of non-convex rotating robots, unlike a closest-point barrier, and enabled docking that bounding circles preclude, the surface cover letting robots pass within the sum of the two certified levels. A Franka Panda whose eight mesh links were constrained against three non-convex obstacles avoided collision in all 30 trials in which the unfiltered motion collides, coming within 1.1 mm, whereas a closest-point barrier collided in 7, and sphere and surface-sample barriers without margins in 17 and 19; two Pandas sharing a workspace kept their meshes apart. The price of the exact safe set is speed near contact and a cost per step that grows with the number of boxes close to contact.

Index Terms—Collision avoidance, control barrier functions, signed distance fields, configuration space, Bernstein polynomials, B-splines, multi-robot systems, manipulators.

I. Introduction

CONTROL barrier functions (CBFs) [1] turn safety conditions into constraints on the control input, typically enforced by a quadratic program (QP) that minimally modifies a nominal command. For collision avoidance, their value is determined by the geometric surrogate behind the barrier. Spheres and super-ellipsoids [2], [3] are smooth but conservative for irregular shapes;

polytopic formulations [4], [5] are exact for convex pieces but become pairwise and piecewise for non-convex bodies; learned, point-cloud, and sampled barriers [6]–[9] trade exactness for generality.

Jo et al. [10] proposed geometry-aware CBFs on Bernstein-polynomial SDFs [11]: the barrier is the robot’s SDF evaluated at the closest point of the obstacle boundary, obtained from the Karush–Kuhn–Tucker (KKT) conditions. That formulation has three limitations. L1) Its time derivative requires the closest point to move smoothly, which holds only for convex level sets. L2) For composite or non-convex geometry, the closest point jumps and the barrier is non-smooth; the remedy of [10], inflating obstacles by the robot’s bounding circle, ignores orientation. L3) The enclosure of the true geometry by the fitted level set rests on a user-chosen margin.

This article removes all three. Its starting point is that two compact bodies can only begin to intersect at a point that lies on both boundaries (Lemma 2). Keeping the SDF of one body above its level on the whole boundary of the other therefore suffices, and no closest point is needed. This condition on the continuous certified boundary is our safe set, which we enforce online: the boundary of one body is covered by small boxes, and on each box a lifted barrier, which adds the body’s own SDF, yields finitely many constraints affine in the inputs that suffice at every point of the box (Fig. 1b). The boxes enter only the constraints on the inputs, never the safe set, and their effect is of second order in the box size. The construction needs one field per body, not per pair, and applies unchanged to planar and spatial bodies and to articulated robots. For planar pairs, whose relative pose is only three-dimensional, the minimum can instead be compiled offline into one smooth field over (x, y, θ) per pair (Fig. 1c); a single scalar barrier then encodes non-convex shapes and rotation, at the price of a level that absorbs the error of the training data.

A. Contributions

- 1) Boundary reduction and certified regions (Sec. IV): a first-contact lemma for compact bodies in $\mathbb{R}^n$, and certified levels of fitted SDFs that enclose the true geometry, for spline boundaries and for triangle meshes (resolves L3).

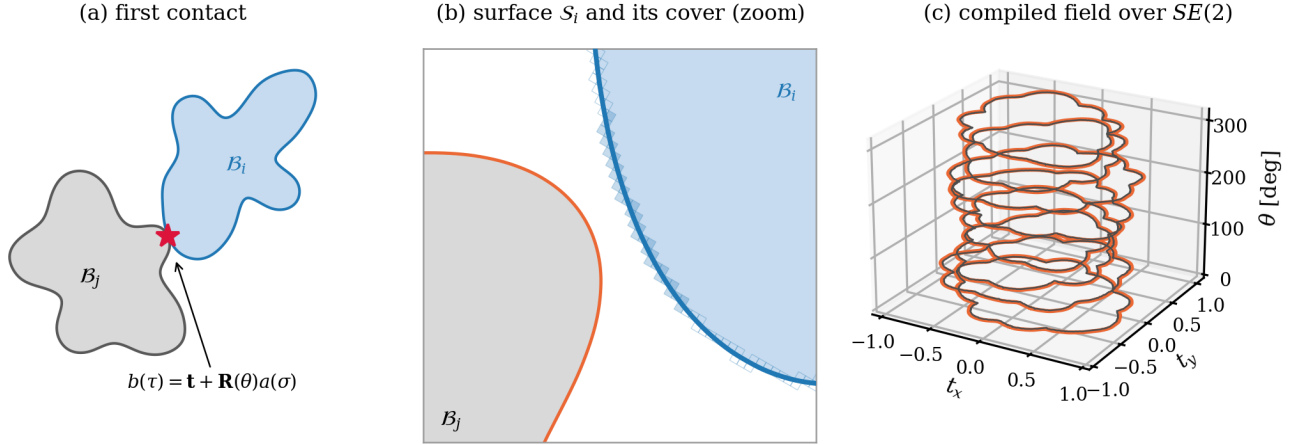


Fig. 1. Both constructions for two planar bodies. (a) Every first contact occurs at a configuration $\mathbf{t} = b(\tau) - \mathbf{R}(\theta)a(\sigma)$ that joins boundary points of both bodies (Lemma 2). (b) Surface cover (Sec. V), with B_i 3 cm before contact: the safe set requires B_j 's SDF to stay above its certified level set (orange) on every point of the continuous certified curve S_i (thick blue); the 5-mm boxes that cover S_i (faint; filled: boxes carrying constraints) only partition it to generate the constraints on the inputs; fills: ground truth. (c) Compiled field (Sec. VI): the configuration-space obstacle C_{ij} (gray: rasterized reference, eight orientations) and the certified level set $\{\phi_{ij} = l_{ij}^*\}$ of one B-spline field over $SE(2)$ (orange).

- 2) Surface-cover barriers (Sec. V): a safe set defined on the continuous level set of each body's SDF, enforced through Bernstein coefficient constraints on a lifted barrier over a Bernstein cover of that level set, with certified Taylor remainders from C^2 derivative majorants; the constraints are affine in the inputs of planar robots and in the joint velocities of manipulators, their conservatism is of second order in the box size and falls on the inputs only, and no closest points or online optimization are needed (resolves L1 and L2).
- 3) Compiled barrier for planar pairs (Sec. VI): one smooth barrier per pair of non-convex rotating bodies over $SE(2)$, whose level is certified on the explicit three-parameter first-contact set by a Bernstein branch-and-bound, together with certificates of its activation band and regularity.
- 4) Evaluation (Sec. VIII): planar docking, multi-robot, and static-obstacle experiments with both constructions under single-integrator and unicycle dynamics against closest-point, bounding-circle, sampling-based, and ball-world baselines, a comparison of Bernstein-form representations, a 7-DOF Franka Panda among non-convex obstacles against sphere and surface-sample barriers, and two Pandas in a shared workspace.

B. Relation to Prior Work

This article keeps the problem formulation, the Bernstein-form distance fields, the pairwise multi-robot QP, and the experimental protocol of [10]. It replaces the closest-point barrier (retained as a baseline), the bounding-circle configuration space, and the heuristic margin. New are the first-contact reduction, both barrier constructions with their certificates and derivatives, the

feasibility result, the representation analysis, and all experiments. The certified enclosure of planar boundaries by B-spline curves and the conversion of B-splines to Bernstein form follow [12] and are not claimed as contributions.

II. Related Work

Geometric barriers. Pairwise barriers between discs or spheres are the standard multi-robot certificates [2], and super-ellipsoids extend them to elongated vehicles [3]; unions of primitives multiply the constraints. For convex polytopes, signed distances through convex programs in the Minkowski-difference space yield exact barriers [4], [13], and smooth barriers exist between strongly convex regions [5]; non-convex bodies must be decomposed. Explicit smooth barriers for polygons compose edge-wise bounds with smooth extrema [14]. Sampled distance functions yield nonsmooth barriers with quantified sampling error [9], and diffeomorphisms to ball worlds handle non-convex unsafe sets without undesired equilibria [15]; we compare with adaptations of both.

Distance fields for manipulators. Bernstein-polynomial distance fields of articulated robots support whole-body collision avoidance, with obstacle points queried against the robot field [11]; neural configuration-space distance functions serve as CBFs with robust margins [16], and differentiable distance fields map obstacle motion into joint space [17]. These works establish distance fields as barriers; they do not certify that the constraint holds on the whole continuous surface of the fitted or true geometry, which is what our surface covers provide.

Configuration-space obstacles and Bernstein bounds. Configuration-space obstacles of polygons can be constructed exactly [18], [19] or by grid convolution [20]; we use the latter only to produce training data. The convex-hull property and de Casteljau subdivision give

rigorous, successively tighter bounds on polynomials over boxes [21], [22]; they were used in [12] to certify planar spline enclosures, and we use them in up to three variables on tensor-product fields.

III. Preliminaries and Problem Statement

CBFs. Consider $\dot{\mathbf{z}} = f(\mathbf{z}) + g(\mathbf{z})\mathbf{u}$ with f, g locally Lipschitz, $\mathbf{u} \in \mathcal{U}$, and a C^1 function h with $\mathcal{H} = \{\mathbf{z} \mid h(\mathbf{z}) \geq 0\}$.

Lemma 1 ([1]). If a locally Lipschitz feedback $\mathbf{k}(\mathbf{z}) \in \mathcal{U}$ satisfies $L_f h + L_g h \mathbf{k} \geq -\gamma h$ with $\gamma > 0$ on an open set containing $\mathcal{H}$, then $\mathcal{H}$ is forward invariant.

Bernstein form. The cubic Bernstein basis $B_r(\xi) = \binom{3}{r} \xi^r (1 - \xi)^{3-r}$ is nonnegative and sums to one, so $p = \sum_r \beta_r B_r$ satisfies $\min_r \beta_r \leq p \leq \max_r \beta_r$ on $[0, 1]$ (convex-hull property), also for tensor products. With M_B mapping Bernstein to power coefficients and $T_{jr} = \binom{r}{j} \xi_0^{r-j} \Delta^j$ ($j \leq r$), the coefficients of p restricted to $[\xi_0, \xi_0 + \Delta]$ and reparametrized to $[0, 1]$ are

$$\beta' = S(\xi_0, \xi_0 + \Delta) \beta, \quad S = M_B^{-1} T(\xi_0, \Delta) M_B, \quad (1)$$

applied along each axis; they converge to the values of p as $\Delta \rightarrow 0$. On each span, a uniform cubic B-spline with control values $\mathbf{P}_c, \dots, \mathbf{P}_{c+3}$ has Bernstein coefficients

$$\beta_c = Q [\mathbf{P}_c, \dots, \mathbf{P}_{c+3}]^\top, \quad Q = \frac{1}{6} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 1 & 4 & 1 \end{bmatrix}, \quad (2)$$

so bounds on a tensor-product spline, and on its derivatives, follow from finitely many coefficients per cell [12], [21].

Problem. Robot bodies and obstacles are compact sets $\mathcal{G}_i \subset \mathbb{R}^n$, $n \in \{2, 3\}$ (the ground truth, not necessarily convex), rigidly attached to frames with poses $(\mathbf{R}_i, \mathbf{p}_i)$ that depend on the state: the pose of a mobile robot, or the forward kinematics of a link. Given nominal inputs, find minimally modified inputs under which no two bodies ever intersect, with constraints whose correctness is certified rather than sampled.

IV. Certified Regions and First Contact

A. First Contact

Lemma 2 (First contact). Let $\mathcal{K}_A(s), \mathcal{K}_B(s) \subset \mathbb{R}^n$, $s \in [0, T]$, be images of compact sets under rigid motions that are continuous in s , disjoint at $s = 0$ and intersecting at $s = T$. Then at $s^* = \inf\{s \mid \mathcal{K}_A(s) \cap \mathcal{K}_B(s) \neq \emptyset\} > 0$, some point lies on both boundaries, $\partial\mathcal{K}_A(s^*) \cap \partial\mathcal{K}_B(s^*) \neq \emptyset$.

Proof. The set of intersecting parameters is closed, so the bodies intersect at $s^* > 0$; pick $\mathbf{z}$ in both. If $\mathbf{z}$ were interior to $\mathcal{K}_B(s^*)$, the point of $\mathcal{K}_A$ that is at $\mathbf{z}$ at s^* would, by continuity, lie in $\mathcal{K}_B(s)$ for $s < s^*$ close to s^* , a contradiction. Symmetrically, $\mathbf{z}$ is not interior to $\mathcal{K}_A(s^*)$, since the point of $\mathcal{K}_B$ at $\mathbf{z}$ would lie in $\mathcal{K}_A$ before s^* . $\square$

The lemma requires neither convexity nor smoothness. It reduces collision avoidance to excluding coincident boundary points, provided that the trajectory starts collision-free.

B. Certified Regions

Each body is replaced by a region $\mathcal{B}_i \supseteq \mathcal{G}_i$ described by a fitted smooth function whose enclosure of $\mathcal{G}_i$ is certified—a sublevel set of an SDF for the surface cover, the inside of a spline curve for the compiled field—and safety for the regions then implies safety for the ground truth.

Level sets of SDFs. Let ϕ_i be a tensor-product cubic B-spline fitted to (approximate) signed distances of $\mathcal{G}_i$ on a box Ω_i , and let $\kappa_i(\mathbf{c})$ be a certified bound on $\|\nabla^2 \phi_i\|$ over the ball of radius $\rho \leq h$ (the cell size) around $\mathbf{c}$: a cubic B-spline whose control value m is the maximum of the cellwise bounds from the Bernstein coefficients of the second derivatives over cells $m - 4, \dots, m + 1$ per axis, which is C^2 and, its weights being a partition of unity, bounds every cell adjacent to that of $\mathbf{c}$. The smallest level that encloses the boundary is $\bar{l}_i = \max_{\partial\mathcal{G}_i} \phi_i$, and we certify it by branch-and-bound over pieces of the boundary: triangles for a mesh, edges for a polygon, and, for a body given by an exact SDF, boxes of an octree that may contain boundary points. For a piece $\mathcal{T}$ with center $\mathbf{c}$ and radius ρ (every point within ρ of $\mathbf{c}$), Taylor's theorem gives

$$\max_{\mathcal{T}} \phi_i \leq \text{UB}(\mathcal{T}) = \phi_i(\mathbf{c}) + \|\nabla \phi_i(\mathbf{c})\| \rho + \frac{1}{2} \kappa_i(\mathbf{c}) \rho^2, \quad (3)$$

and ϕ_i at vertices, centers, or projected points on $\partial\mathcal{G}_i$ gives attained values whose maximum ℓ lower-bounds $\bar{l}_i$. Pieces with $\text{UB} < \ell + \epsilon$ are discarded; the others are split (a triangle into four, an edge or a box into halves or octants) and re-bounded.

Proposition 1. When no piece is left, $l_i = \ell + \epsilon$ satisfies $\partial\mathcal{G}_i \subset \{\phi_i < l_i\}$ and $l_i \leq \bar{l}_i + \epsilon$. The procedure terminates for every $\epsilon > 0$.

Proof. Every boundary point lies in a discarded piece, where $\phi_i \leq \text{UB} < \ell + \epsilon$ with ℓ nondecreasing, and $\ell \leq \bar{l}_i$ because it is attained. Splitting halves ρ , and $\text{UB} - \phi_i(\mathbf{c}) \leq G\rho + \frac{1}{2} \kappa \rho^2 \rightarrow 0$ with $\phi_i(\mathbf{c}) \leq \ell$ for centers on the boundary (for octree boxes, $\phi_i(\mathbf{c}) \leq \bar{l}_i + G\rho$), so every piece is discarded after finitely many splits. $\square$

The certificate is on the whole continuous boundary, is tight up to ϵ , and needs no sampling or distance computation; the fit affects only the size of l_i , never its validity, so the training targets may be approximate. For the surface cover, $\mathcal{B}_i = \{\phi_i \leq l_i\}$, and we write $\mathcal{S}_i = \{\phi_i = l_i\}$ for the certified surface.

Planar spline boundaries [12], used by the compiled field. The boundary is a closed periodic cubic B-spline $a_i : \mathbb{S}^1 \rightarrow \mathbb{R}^2$, fitted to outward-offset samples of $\mathcal{G}_i$ and accepted only if every Bézier control hull (after subdivision) misses $\mathcal{G}_i$ and the curve winds around a point of $\mathcal{G}_i$. The region $\mathcal{B}_i$ is the curve with the points of nonzero winding number; for connected $\mathcal{G}_i$, $\mathcal{G}_i \subseteq \mathcal{B}_i$ and $\partial\mathcal{B}_i \subseteq a_i(\mathbb{S}^1)$.

V. Surface-Cover Barriers

We keep one SDF per body and enforce the first-contact condition on the whole certified surface; the construction is

the same for planar and spatial bodies, rigid or articulated. Let body A (e.g., a robot or a link) have the certified surface $\mathcal{S}_A = \{\phi_A = l_A\}$ in its frame, and let body B (e.g., another robot or an obstacle) have the field ϕ_B and level l_B in its frame, both from Proposition 1. We write $\mathbf{y}(\mathbf{x})$ for a point $\mathbf{x}$ of A mapped into B 's frame by the current relative pose. For planar bodies, the fields are bivariate and the boxes below are squares.

A. Safe Set on the Continuous Surface

The safe set is the first-contact condition on every point of the certified surface,

$$\mathcal{H}_{AB} = \{\text{poses} \mid h(\mathbf{x}) = \phi_B(\mathbf{y}(\mathbf{x})) - l_B \geq 0 \quad \forall \mathbf{x} \in \mathcal{S}_A\}; \quad (4)$$

no box, sample, or bound enters its definition.

Theorem 1. Suppose $\{\phi_A \leq l_A\}$ and $\{\phi_B \leq l_B\}$ are compact subsets of their domains, the bodies start with $\{\phi_A \leq l_A\} \cap \{\phi_B < l_B\} = \emptyset$, and a continuous trajectory stays in $\mathcal{H}_{AB}$. Then $\mathcal{G}_A$ and $\mathcal{G}_B$ never intersect.

Proof. For $\epsilon > 0$, if $\{\phi_A \leq l_A\}$ met $\{\phi_B \leq l_B - \epsilon\}$, Lemma 2 would give a first point on both boundaries, where $\phi_A = l_A$ and $\phi_B = l_B - \epsilon$, contradicting $h \geq 0$ on $\mathcal{S}_A$. Hence $\{\phi_A \leq l_A\} \cap \{\phi_B < l_B\} = \emptyset$ throughout. A first contact of $\mathcal{G}_A$ and $\mathcal{G}_B$ would lie on $\partial\mathcal{G}_A \cap \partial\mathcal{G}_B$ (Lemma 2), where $\phi_A < l_A$ and $\phi_B < l_B$ by Proposition 1. $\square$

B. Certified Cover and Lifted Barrier

On each cell of ϕ_A , the Bernstein coefficients of $\phi_A - l_A$ follow from (2). If they all have one strict sign on a box, the box contains no point of $\mathcal{S}_A$ by the convex-hull property. Starting from the cells and halving every box whose coefficients change sign, restricted by (1), until its side is at most $s_{\max}$, yields finitely many boxes $\{\mathcal{Q}_k\}$ of side s , with centers $\mathbf{c}_k$ and half-diagonal r , whose union contains all of $\mathcal{S}_A$, not a sample of it. The cover is computed once per body; it only partitions the surface for the enforcement below and does not enter (4).

By the comparison lemma, $\mathcal{H}_{AB}$ stays invariant if $\tilde{h}(\mathbf{x}) \geq -\gamma h(\mathbf{x})$ at every surface point. The surface is implicit, so we impose this condition on whole boxes, after lifting the barrier so that the points of a box off the surface do not tighten it: for $\mathbf{x}$ in a box,

$$\tilde{h}(\mathbf{x}) = \phi_A(\mathbf{x}) - l_A + \phi_B(\mathbf{y}(\mathbf{x})) - l_B, \quad (5)$$

which equals h on $\mathcal{S}_A$. Since $\mathbf{x}$ is fixed in A 's frame, $\phi_A(\mathbf{x})$ does not change, and $\tilde{h}(\mathbf{x}) = \nabla\phi_B(\mathbf{y})^\top \dot{\mathbf{y}}(\mathbf{x})$. A box point displaced from the surface by δ toward B has $\phi_A - l_A \approx \delta$ and $\phi_B - l_B \approx h - \delta$ for SDF-like fields, so $\tilde{h} \approx h$: the lifting cancels the first-order effect of the box size. The box points move with $\dot{\mathbf{y}}(\mathbf{x}) = \mathbf{J}\mathbf{u} + \mathbf{W}(\mathbf{u})(\mathbf{y} - \mathbf{y}_c)$, affine in the inputs $\mathbf{u}$, where $\mathbf{J}\mathbf{u}$ is the velocity of the box center in B 's frame and $\mathbf{W}(\mathbf{u})$ the relative angular velocity as a skew matrix (Sec. V-D).

C. Bernstein Coefficient Constraints

Cellwise bounds on the Frobenius norm of the third-derivative tensor follow from the Bernstein coefficients of the third partial derivatives. On each (fixed) box of A , M_A is their maximum over the at most 2^n cells meeting the box. For ϕ_B , the bound must vary continuously with the pose: M_B is piecewise linear on a grid of spacing $h/2$, with node values the maxima over the cells meeting the cube of half-width $h/2 + r$ around the node; interpolation is a convex combination of nodes whose cubes all contain the ball of radius r around $\mathbf{y}_c$, so M_B bounds the tensor on that ball and is Lipschitz. Let q_A and q_B be the second-order Taylor polynomials of ϕ_A at $\mathbf{c}$ and of ϕ_B at $\mathbf{y}_c$. Taylor's theorem gives, on a box, $\phi_A \geq q_A - \frac{1}{6}M_A r^3$, $\phi_B \geq q_B - \frac{1}{6}M_B r^3$, and $\|\nabla\phi_B - \nabla q_B\| \leq \frac{1}{2}M_B r^2$. The box points move relative to B with the twist $(\mathbf{V}, \mathbf{\Omega})$ of A relative to B at A 's origin $\mathbf{p}_A$, linear in $\mathbf{u}$, so $\|\dot{\mathbf{y}}(\mathbf{x})\| \leq \|\mathbf{V}\|_1 + \rho\|\mathbf{\Omega}\|_1$ with $\rho = \|\mathbf{c} - \mathbf{p}_A\| + r$. With $\mathbf{d} = \mathbf{y} - \mathbf{y}_c$, $\|\mathbf{d}\| \leq r$, $\mathbf{g} = \nabla\phi_B(\mathbf{y}_c)$, $\mathbf{H} = \nabla^2\phi_B(\mathbf{y}_c)$, and $\dot{\mathbf{y}}_c = \mathbf{J}\mathbf{u}$, the models give on the box

$$\begin{aligned} \Psi \geq & \mathbf{g}^\top \dot{\mathbf{y}}_c + \mathbf{d}^\top \mathbf{H} \dot{\mathbf{y}}_c + \mathbf{g}^\top \mathbf{W} \mathbf{d} - r^2 \|\mathbf{H}\| \|\mathbf{\Omega}\| \\ & + \gamma(q_A - l_A + q_B - l_B - \frac{1}{6}(M_A + M_B)r^3) \\ & - \frac{1}{2}M_B r^2 (\|\mathbf{V}\|_1 + \rho\|\mathbf{\Omega}\|_1), \end{aligned} \quad (6)$$

where $\Psi(\mathbf{x}, \mathbf{u}) = \nabla\phi_B(\mathbf{y})^\top \dot{\mathbf{y}}(\mathbf{x}) + \gamma\tilde{h}(\mathbf{x})$; the first three terms are affine in $\mathbf{d}$ and in $\mathbf{u}$.

Proposition 2. Let β_k , $k = 1, \dots, 3^n$, be the Bernstein coefficients of degree two per coordinate of $q_A - l_A + q_B(\mathbf{y}(\cdot)) - l_B$ on a box $\mathcal{Q}$, let $\mathbf{d}_j$, $j = 1, \dots, 2^n$, be the vertices of $\mathcal{Q}$ relative to its center in B 's frame, and let $\sigma_V = \frac{1}{2}M_B r^2$ and $\sigma_\Omega = \frac{1}{2}\rho M_B r^2 + r^2 \|\mathbf{H}\|$. If, for all j ,

$$\begin{aligned} & (\mathbf{g} + \mathbf{H}\mathbf{d}_j)^\top \mathbf{J}\mathbf{u} + \mathbf{g}^\top \mathbf{W}(\mathbf{u})\mathbf{d}_j - \sigma_V \mathbf{1}^\top \mathbf{a}_V - \sigma_\Omega \mathbf{1}^\top \mathbf{a}_\Omega \\ & + \gamma(\min_k \beta_k - \frac{1}{6}(M_A + M_B)r^3) \geq 0, \end{aligned} \quad (7)$$

$$\mathbf{a} = (\mathbf{a}_V, \mathbf{a}_\Omega) \geq |(\mathbf{V}, \mathbf{\Omega})|,$$

then $\Psi(\mathbf{x}, \mathbf{u}) \geq 0$ for every $\mathbf{x} \in \mathcal{Q}$.

Proof. In (6), the terms affine in $\mathbf{d}$ attain their minimum over the (rotated) box at a vertex, and the value model is at least $\min_k \beta_k$ by the convex-hull property; the twist terms are bounded by $\mathbf{a}$. $\square$

The constraints, 2^n per box, are affine in $(\mathbf{u}, \mathbf{a})$. The epigraph variables $\mathbf{a}$ (six per pair of bodies in space, three in the plane) are shared by all boxes of the pair, and since the left side of (7) decreases in $\mathbf{a}$, the feasible set in $\mathbf{u}$ is exactly that of the same constraints with $|(\mathbf{V}, \mathbf{\Omega})|$; the filter remains a QP. Joint-by-joint speed bounds would be 3 to 10 times larger in Sec. VIII-F, since joint velocities partly cancel. The remainder terms tighten only the admissible inputs, and the one proportional to the twist vanishes at rest. Online, ϕ_B , its first two derivatives, and M_B are evaluated at box centers; the Taylor data of ϕ_A are computed offline. At rest, the loss of the constraints is of second order in the box size where the two level sets face each other:

Proposition 3. With $\mathbf{b} = \nabla\phi_A(\mathbf{c}) + \mathbf{R}^\top \nabla\phi_B(\mathbf{y}_c)$ and $\mathbf{Q} = \nabla^2\phi_A(\mathbf{c}) + \mathbf{R}^\top \nabla^2\phi_B(\mathbf{y}_c)\mathbf{R}$, where $\mathbf{R}$ maps A 's frame to B 's, the constant terms of (7) are at least $\gamma(\tilde{h}(\mathbf{c}) - \frac{s}{2}\|\mathbf{b}\|_1 - \frac{s^2}{8}\sum_{ij}|Q_{ij}| - \frac{1}{6}(M_A + M_B)r^3)$.

Proof. With $\mathbf{x} = \mathbf{c} + s\boldsymbol{\tau}$, $\boldsymbol{\tau} \in [-\frac{1}{2}, \frac{1}{2}]^n$, the degree-two coefficients of τ_i , $\tau_i\tau_j$, and τ_i^2 lie in $[-\frac{1}{2}, \frac{1}{2}]$, $[-\frac{1}{4}, \frac{1}{4}]$, and $[-\frac{1}{4}, \frac{1}{4}]$. $\square$

Near a contact of SDF-like fields, $\nabla\phi_A \approx -\mathbf{R}^\top \nabla\phi_B$, so $\|\mathbf{b}\|$ is of the order of the curvatures times r and the whole loss is of order r^2 ; a bound on ϕ_B alone over a box loses $\|\nabla\phi_B\|r$ instead.

D. Derivatives and Pruning

For planar $SE(2)$ single integrators, a point of A at world position $\mathbf{w}$ moves in B 's frame with $\mathbf{R}(\theta_B)^\top(\mathbf{v}_A + \omega_A J(\mathbf{w} - \mathbf{p}_A) - \mathbf{v}_B - \omega_B J(\mathbf{w} - \mathbf{p}_B))$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, and $\mathbf{W} = (\omega_A - \omega_B)J$. For a link A of an articulated robot, a revolute joint j before the link contributes the column $\mathbf{z}_j \times (\mathbf{w} - \mathbf{o}_j)$ and the angular velocity $\mathbf{z}_j \dot{q}_j$; for two moving arms, the joints of B 's arm enter with the opposite sign. In every case, $\mathbf{J}$ and $\mathbf{W}$ are linear in the inputs, and no closest point and no optimization are involved.

Pruning. A box needs no constraint if the constant terms of (7), divided by γ , are at least an activation threshold $\eta > 0$: then $h \geq \eta$ on its part of the surface. We group the boxes of each body into clusters with center $\bar{\mathbf{c}}$ and radius R (covering the boxes' balls) and skip a cluster if $\phi_B(\bar{\mathbf{c}}) - GR - l_B + \min(\phi_A - l_A) \geq \eta$, where G bounds $\|\nabla\phi_B\|$ on the cells the cluster can reach and the minimum of $\phi_A - l_A$ over the cluster's boxes is certified offline; remaining boxes are first screened with $\phi_B(\mathbf{y}_c) - \|\nabla\phi_B(\mathbf{y}_c)\|r - \frac{1}{2}\kappa_B(\mathbf{y}_c)r^2 - l_B + \min_Q(\phi_A - l_A) \geq \eta$.

Refinement. Near sharp edges, $\frac{1}{2}M_B r^2$ can exceed $\|\nabla\phi_B\|$: the gradient direction is undetermined on the box, no motion can be certified, and the bodies freeze. An active box with $\frac{1}{2}M_B r^2 > \vartheta$ is therefore replaced online by those of its 2^n half-size boxes whose bounds of $\phi_A - l_A$ change sign; they cover the same part of $\mathcal{S}_A$ and quarter both remainders. We use $\vartheta = 0.1$, at most two halvings, and only boxes whose lower bound of $\tilde{h}$ is below 1 cm.

Theorem 2. Let a locally Lipschitz feedback satisfy (7) on every box, or on the kept half-size boxes of a refined box, whose certified lower bound of $\tilde{h}$ is below η . Then $\mathcal{H}_{AB}$ is forward invariant.

Proof. Fix $\mathbf{x} \in \mathcal{S}_A$ with $h(\mathbf{x}) \geq 0$ initially. Whenever $h(\mathbf{x}) < \eta$, every box containing $\mathbf{x}$ has a lower bound of $\tilde{h}$ below $\tilde{h}(\mathbf{x}) = h(\mathbf{x}) < \eta$, so it is constrained, and Proposition 2 gives $\dot{h}(\mathbf{x}) = \Psi(\mathbf{x}, \mathbf{u}) - \gamma h(\mathbf{x}) \geq -\gamma h(\mathbf{x})$. Hence $h(\mathbf{x})$ cannot cross zero. $\square$

VI. Compiling a Planar Pair into One Barrier

The surface cover enforces the first-contact condition through many coefficient constraints. For planar bodies, whose relative pose is only three-dimensional, the same condition can instead be compiled offline into one smooth

field over $SE(2)$ per pair. This yields a single barrier per pair and a per-step cost that is independent of the geometry; its level is certified directly on the explicit first-contact set.

A. Configuration-Space Field and Barrier

For planar bodies with poses $(\mathbf{p}_i, \theta_i)$, the configuration of robot i relative to robot j is $\mathbf{q}_{ij} = (\mathbf{t}_{ij}, \theta_{ij})$ with $\mathbf{t}_{ij} = \mathbf{R}(-\theta_j)(\mathbf{p}_i - \mathbf{p}_j)$ and $\theta_{ij} = \theta_i - \theta_j$, and the robots intersect if and only if $\mathbf{q}_{ij}$ lies in the configuration-space obstacle $\mathcal{C}_{ij} = \{(\mathbf{t}, \theta) \mid (\mathbf{R}(\theta)\mathcal{B}_i + \mathbf{t}) \cap \mathcal{B}_j \neq \emptyset\}$. We fit, offline, a field on $\Omega_{ij} = [-D_{ij}, D_{ij}]^2 \times \mathbb{S}^1$ that approximates the signed distance to $\mathcal{C}_{ij}$ (negative inside),

$$\phi_{ij}(\mathbf{t}, \theta) = \sum_{a,b,c} W_{abc} N_a(t_x) N_b(t_y) \tilde{N}_c(\theta), \quad (8)$$

with uniform cubic B-spline bases and a periodic basis $\tilde{N}_c$ in θ , and $D_{ij} > \rho_i + \rho_j$, where ρ_i bounds the radius of $\mathcal{B}_i$. Given a certified level l_{ij}^* (Sec. VI-B), the barrier is

$$h_{ij} = \phi_{ij}(\mathbf{q}_{ij}) - l_{ij}^*, \quad (9)$$

which is C^2 in the poses. As training data, we rasterize $\mathcal{C}_{ij}$ at 96 orientations by convolving occupancy masks, take Euclidean distance transforms, and fit W by separable ridge regression; the certificate never samples orientations.

Proposition 4. With $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, $\mathbf{g}_t = \nabla_{\mathbf{t}}\phi_{ij}$, $g_\theta = \partial_\theta\phi_{ij}$, and $SE(2)$ single integrators $\dot{\mathbf{p}}_i = \mathbf{v}_i$, $\dot{\theta}_i = \omega_i$,

$$\dot{h}_{ij} = (\mathbf{R}(\theta_j)\mathbf{g}_t)^\top(\mathbf{v}_i - \mathbf{v}_j) + g_\theta\omega_i - (\mathbf{g}_t^\top J\mathbf{t}_{ij} + g_\theta)\omega_j. \quad (10)$$

Proof. $\dot{\mathbf{t}}_{ij} = -\omega_j J\mathbf{t}_{ij} + \mathbf{R}(-\theta_j)(\mathbf{v}_i - \mathbf{v}_j)$ and $\dot{\theta}_{ij} = \omega_i - \omega_j$; apply the chain rule. $\square$

For unicycles, substituting $\mathbf{v}_k = v_k[\cos\theta_k, \sin\theta_k]^\top$ keeps (10) affine in the inputs; we monitor the norm of its coefficient vector in the experiments.

For a robot among static obstacles, the configuration-space obstacle of a union is the union of the configuration-space obstacles, so one field over the workspace and θ encodes all obstacles in a single barrier whose online cost is independent of their number.

B. Contact Certificate

By Lemma 2 applied to $\mathcal{B}_i$ and $\mathcal{B}_j$ with $\partial\mathcal{B} \subseteq a(\mathbb{S}^1)$, every first contact occurs in

$$\mathcal{M}_{ij} = \{(b(\tau) - \mathbf{R}(\theta)a(\sigma), \theta) \mid \sigma, \tau, \theta \in \mathbb{S}^1\}, \quad (11)$$

an explicit three-parameter family; the configuration-space obstacle, which would require a Minkowski sum for every orientation, is never formed.

Definition 1. A level l_{ij}^* is a contact certificate if $\mathcal{M}_{ij} \subset \Omega_{ij}$ and $\phi_{ij} < l_{ij}^*$ on $\mathcal{M}_{ij}$.

Theorem 3. Let l_{ij}^* be a contact certificate, and let a continuous trajectory start with $\mathbf{q}_{ij}(0) \notin \mathcal{C}_{ij}$ and satisfy $h_{ij} \geq 0$ whenever $\mathbf{q}_{ij} \in \Omega_{ij}$. Then $\mathcal{B}_i$ and $\mathcal{B}_j$, and hence $\mathcal{G}_i$ and $\mathcal{G}_j$, never intersect.

Algorithm 1 Contact certificate for a pair (i, j)

Require: spans of a and b , field ϕ_{ij} , tolerance $\varepsilon > 0$

- 1: $\mathcal{Q} \leftarrow$ all span pairs $\times n_\theta$ angle intervals; $\ell \leftarrow \max_{\mathcal{Q}} \text{LB}$
- 2: while $\mathcal{Q} \neq \emptyset$ do
- 3: remove every $\mathcal{R}$ with $\text{UB}(\mathcal{R}) < \ell + \varepsilon$
- 4: bisect each remaining $\mathcal{R}$ along its widest parameter image
- 5: $\ell \leftarrow \max(\ell, \max_{\mathcal{Q}} \text{LB})$
- 6: end while
- 7: return $l_{ij}^* = \ell + \varepsilon$

Proof. A first contact would lie in $\mathcal{M}_{ij} \subset \Omega_{ij}$, where $h_{ij} < 0$. $\square$

Branch-and-bound. After (2), both boundaries are piecewise cubic Bézier curves. A region $\mathcal{R}$ is a span of a with a parameter interval, a span of b with an interval, and an angle interval. By (1), the sub-curves lie in the hulls of their restricted control points; each rotated control point of a traces an arc whose bounding box is exact, so $\mathbf{t}$ ranges over a box $[\underline{\mathbf{b}} - \bar{\mathbf{a}}, \bar{\mathbf{b}} - \underline{\mathbf{a}}]$ on the part of $\mathcal{M}_{ij}$ covered by $\mathcal{R}$. The maximum of the Bernstein coefficients of ϕ_{ij} restricted to this box and angle interval is an upper bound $\text{UB}(\mathcal{R})$ ($+\infty$ if the box leaves Ω_{ij}); the value at the midpoint parameters is attained on $\mathcal{M}_{ij}$ and is a lower bound $\text{LB}(\mathcal{R})$ of $\max_{\mathcal{M}_{ij}} \phi_{ij}$.

Proposition 5. If Algorithm 1 terminates, l_{ij}^* is a contact certificate and $l_{ij}^* - \max_{\mathcal{M}_{ij}} \phi_{ij} \leq \varepsilon$. It terminates for every $\varepsilon > 0$ whenever $\mathcal{M}_{ij}$ lies in the interior of Ω_{ij} .

Proof. Every point of $\mathcal{M}_{ij}$ lies in a region removed with $\text{UB} < \ell + \varepsilon \leq l_{ij}^*$ (ℓ is nondecreasing), and a finite UB requires the box to lie in Ω_{ij} ; ℓ is attained on $\mathcal{M}_{ij}$. For termination, the control-hull diameters of the sub-curves are bounded by the curve derivatives times the parameter lengths, so the image widths of any nested chain of surviving regions tend to zero; the restricted coefficients of a cubic cell differ from its values by at most a constant times the box diameter, so $\text{UB} - \text{LB} \rightarrow 0$ uniformly, and a region survives only while $\text{UB} - \text{LB} \geq \varepsilon$. $\square$

Termination also certifies $\mathcal{M}_{ij} \subset \Omega_{ij}$, so the radius bounds ρ_i affect only efficiency. Two further certificates use the same bounds. The activation band: $h_{ij} > 0$ wherever $D_{ij} - \delta \leq \|\mathbf{t}\|_\infty \leq D_{ij}$, so a pair enters the QP with a positive barrier; the minimum restricted coefficient of every cell meeting the band must exceed l_{ij}^* . Regularity: bisecting every cell whose coefficient range contains l_{ij}^* until either l_{ij}^* leaves the range or the coefficients of one partial derivative have one strict sign certifies $\nabla \phi_{ij} \neq 0$ on $\{h_{ij} = 0\}$, so the barrier row never vanishes there. All certificates assume exact arithmetic; our implementation uses double precision without outward rounding.

C. Representations

The global Bernstein polynomial of [10], piecewise Bézier patches, and cubic B-splines are all cellwise Bern-

stein forms, so Algorithm 1 applies to each. Patches that share only boundary coefficients are C^0 : the constraint normal jumps across cell faces, which violates the C^1 requirement of Lemma 1 and appears as chattering and persistent violations (Sec. VIII-C). The cubic B-spline is C^2 with about a third as many coefficients per cell; the global polynomial is smooth but, being one patch, cannot localize sharp non-convex features.

VII. Safety Filter

Collecting the constraints of all pairs—the constraints (7) of every active box, and one barrier per active pair with a compiled field, gathered in $\mathcal{A}$ —the filter is

$$\begin{aligned}
 (\mathbf{u}^*, \mathbf{a}^*) &= \arg \min_{\mathbf{u} \in \mathcal{U}, \mathbf{a}} \|\mathbf{u} - \mathbf{u}^{\text{nom}}\|^2 + \varepsilon_a \|\mathbf{a}\|^2 \\
 &\text{s.t. (7) for all active boxes,} \\
 &\quad \dot{h}(\mathbf{u}) \geq -\gamma h \quad \forall h \in \mathcal{A},
 \end{aligned} \tag{12}$$

with $\mathbf{u}$ the stacked planar velocities or the joint velocities, $\mathbf{a}$ the epigraph variables of all active pairs, and $\varepsilon_a = 10^{-3}$. With polytopic input bounds, (12) is a QP, which we solve exactly as a least-distance program by non-negative least squares [23], warm-started by constraint generation.

Proposition 6 (Feasibility). For driftless dynamics—planar single integrators, unicycles, or velocity-controlled joints—and any $\mathcal{U} \ni \mathbf{0}$, $(\mathbf{u}, \mathbf{a}) = \mathbf{0}$ is feasible for (12) whenever $h \geq 0$ for all $h \in \mathcal{A}$ and the constant terms of (7) are nonnegative on all active boxes; by Proposition 3, the latter holds wherever $\tilde{h}$ at the box centers exceeds a bound of second order in the box size.

Proof. All barrier derivatives and coefficients are linear in $\mathbf{u}$ without drift, so $\mathbf{u} = \mathbf{0}$ gives $\dot{h} = 0 \geq -\gamma h$ and leaves the constant terms of (7). $\square$

Corollary 1. If the bodies start in the safe sets, constraints enter the QP only with positive values (activation band or pruning threshold), (12) stays feasible, and its minimizer is locally Lipschitz, then in continuous time no pair of bodies, and no pair of ground-truth shapes, ever intersects.

Proof. Lemma 1 keeps the compiled barriers nonnegative and Theorem 2 keeps each pose in $\mathcal{H}_{AB}$; Theorems 3 and 1 conclude. $\square$

The QP is thus feasible on the whole safe set of the compiled field, and on that of the surface cover except in a band of second order in the box size at its boundary, before which the filter may stop the bodies; the safe set itself is unaffected. Feasibility does not imply progress: exact non-convex geometry can hook and block bodies [24], and we do not resolve such deadlocks. In the experiments, a slack variable is added only if (12) is infeasible.

VIII. Experiments

A. Planar Setup

Robot A is a C-shaped body and robot B a three-lobed body with smooth periodic B-spline boundaries

($\rho_A = 0.529$ m, $\rho_B = 0.579$ m), which also serve as ground truth. Control uses $SE(2)$ single integrators with $\gamma = 5$, $v_{\max} = 1$ m/s, $\omega_{\max} = 2$ rad/s, a 10-ms step, and a proportional nominal controller. Unless stated otherwise, the surface cover uses for each body a bivariate cubic B-spline with 1-cm cells fitted to the signed distance of its ground-truth polygon, the tightest level certified on the polygon by Proposition 1 ($\epsilon = 0.1$ mm), 5-mm boxes with four constraints (7) each, and $\eta = 5$ cm; the compiled pair field is a cubic B-spline with $45 \times 45 \times 48$ cells (1.1×10^5 coefficients), certified by Algorithm 1 with $n_\theta = 32$ and $\epsilon = 2$ mm. Computations run on an AMD Ryzen 9 3900X CPU with an NVIDIA RTX 4070 GPU (fitting and certification in double precision on the GPU; control in Python on one CPU thread). Ground-truth shapes are dense polygons; collisions are checked at every step and, between steps, along the interpolated poses with a rigid-motion bound and bisection down to 0.1 mm.

Baselines. The closest-point barrier is the signed distance between the fitted bodies, sampled as polygons, minus a margin $m = 2$ cm, with the derivative at one closest pair; it removes the fitting error of [10] but keeps its single-closest-point structure. The circle barrier [2] uses the minimum enclosing circle of each fitted body, with body-frame center, and the same margin.

B. Representations and Conservatism of the Compiled Field

Table I compares representations of the compiled pair field with about 1.1×10^5 coefficients. Both piecewise fields are certified, including band and regularity; the B-spline level is 14% lower than the C^0 level, and its gradient is continuous to numerical precision. The global polynomial of [10] is 2.8 times less accurate near the boundary, and its sampled maximum on $\mathcal{M}$ already exceeds the certified B-spline level, so no certificate for it could be less conservative. The C^2 requirement applies equally to the body fields of the surface cover, whose constraints (7) contain the gradient and Hessian of ϕ_B and its derivative majorant.

Beyond moderate model sizes, the certified level is set by the training targets rather than by the representation: coarsening the raster from 1 to 2 cm doubles l^* (38.2 mm) and refining it to 0.5 cm halves it (10.7 mm), whereas refining the cells from 8.3 to 4.7 cm leaves it at 18.6–18.7 mm. Only 3.5% of the 38,712 cells met by $\mathcal{M}$ have bounds above $l^*/2$, so a spatially varying level could reduce the conservatism considerably.

C. Docking into a Tapered Receptacle

A fixed module carries two panel arrays and, in its front face, a tapered receptacle; a vehicle has a hull, two nozzles, two panel arrays, and a 0.70-m nose that narrows from 0.32 to 0.10 m (Fig. 2). The receptacle is the seated nose offset outward by 32 mm everywhere, so the nose seats fully (0.668 m past the mouth) only if the barrier admits configurations within 32 mm of contact, and only within

TABLE I
Representations on the C-Shape/Three-Lobed Pair

	B-spline (C^2)	Bézier (C^0)	global $Q=48$	Bernstein $Q=23$ [10]
coefficients [10^3]	111	115	111	12
RMSE near boundary [mm]	2.6	2.6	6.1	9.6
max. error near bd. [mm]	17.0	21.2	48.4	79.8
sampled $\max_{\mathcal{M}} \phi$ [mm]	16.7	17.1	23.3	34.1
certified l^* [mm]	19.9	23.2	—	—
certification time [s]	3.6	2.0	—	—
max. gradient jump	9×10^{-6}	1.69	0	0

Errors: against the reference signed distance d at 16 held-out orientations, where $|d| < 0.1$ m. $\max_{\mathcal{M}} \phi$ is sampled at 2×10^5 points and lower-bounds any valid level; the global polynomials were not certified.

TABLE II
Docking with Different Barriers

	ours			closest	
	surface cover	compiled (B-spline)	compiled (C^0 Bézier)	point	circle
certified level [mm]	1.2, 0.5	12.3	19.0	—	—
nose depth [m]	0.698	0.666	0.692	0.671	−0.084
min. true gap [mm]	1.7	24.8	6.1	28.4	102.5
min. h [mm]	0.0	0.0	−15.8	−0.9	—
input variation	6.4	3.6	161	3.8	2.5
constraints, max.	1604	1	1	1	1
time per step [ms]	6.2	0.48	0.49	6.5	0.22

Nose depth: tip past the mouth at the end (full seating with a 32-mm gap: 0.668 m). Cover: levels of the module and vehicle fields, 1548 boxes of 5 mm, at most 401 active; h : certified lower bound of h . Compiled fields: $82^2 \times 384$ and 30^3 cells. Time: median, complete filter, one CPU thread. The circle barrier's h is in m^2 and stayed nonnegative.

a narrow band of orientations. For the surface cover, the module and vehicle fields have 1.25- and 1-cm cells, their tightest levels certified on the ground-truth polygons are 1.2 and 0.5 mm, and 1548 boxes of 5 mm cover the vehicle's certified curve. For the compiled field, the certified boundaries need offsets of 4 and 5 mm; training targets come from a 0.25-cm raster at 576 orientations; the field has $82 \times 82 \times 384$ cells (2.8×10^6 coefficients), and Algorithm 1 certifies $l^* = 12.3$ mm in 34 s. The vehicle starts 3.4 m away, rotated by 40° , and its nominal goal lies 10 cm past full seating.

With the surface cover, the vehicle docks without violating its barrier and without slack (Fig. 2a, Table II): it rotates while approaching, and its nose passes the full-seating depth by 3.0 cm, closing the 32-mm floor gap to 1.7 mm. This is the sum of the two certified levels (1.2 and 0.5 mm): the 5-mm boxes cost no visible clearance, as Proposition 3 predicts, and halving them to 2.5 mm changes the gap by 0.1 mm. Barriers at the box corners that bound ϕ_B alone, whose safe set depends on the boxes, stop the nose at 6.8 mm (3.9 mm with 2.5-mm boxes). The price is up to 401 active boxes with 1604 constraints.

The compiled field docks as smoothly with a single barrier and an order of magnitude less time per step, but its level absorbs the error of the rasterized training targets

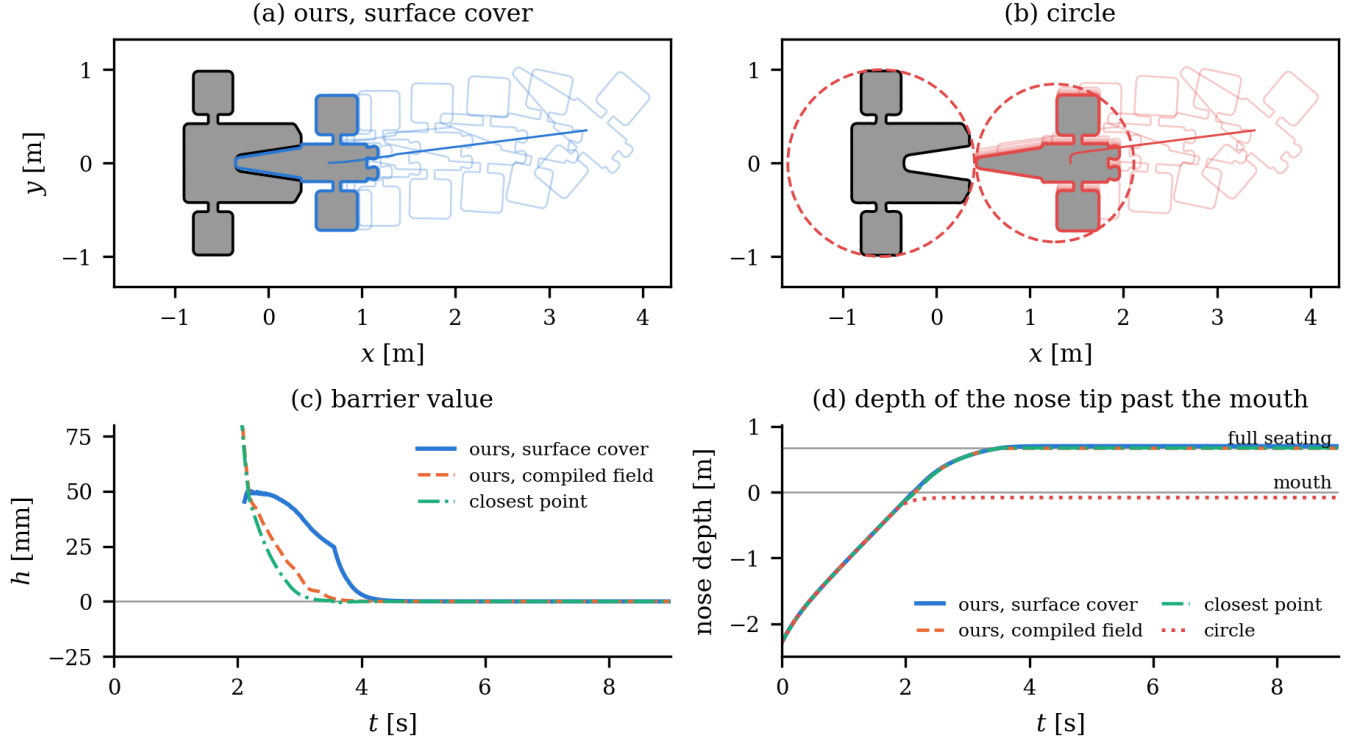


Fig. 2. Docking into a tapered receptacle with the module fixed. (a) Surface-cover barriers: the vehicle rotates while approaching and seats its 0.70-m nose 3.0 cm past full seating, 1.7 mm from the receptacle floor. (b) Minimum enclosing circles (dashed): the vehicle stops 8 cm in front of the mouth. Poses every 0.6 s; gray: ground truth; colored and black: certified boundaries. (c) Barrier value (for the cover, the certified lower bound of $\bar{h}$ over the active boxes): both constructions keep $h \geq 0$, the closest-point barrier violates its constraint by 0.9 mm. (d) Depth of the nose tip past the mouth (full seating: 0.668 m).

(Sec. VIII-B): the nose stops 2 mm short of full seating at $h = 0$, with a ground-truth gap of 24.8 mm. Both C^0 fields push the nose deeper only by violating their constraints by 15.8 and 9.6 mm, with the constraint normal switching across cell faces and the input chattering (total variation 161 and 94.5, against 3.6). The closest-point barrier docks as smoothly here, because the contact stays on the tapered walls and the closest pair does not jump, but costs 13 times more per step than the compiled field; the circles stop the nose 8 cm in front of the mouth. Refining the step from 10 to 1 ms leaves the compiled barrier nonnegative and the final pose unchanged, whereas the violation of the C^0 field stays at 15–16 mm: it is caused by gradient jumps, not by discretization.

D. Multiple Robots

Five shapes are generated as star-shaped radial Fourier series with hull-area ratios below 0.9, and scaled to radii of 0.85–1.08 m. The surface cover needs one field per body: the five tightest levels (0.29–0.35 mm) and covers (1240–1426 boxes), with the Taylor data of the boxes, take 50 s in total, an offline cost linear in the number of bodies. The compiled construction needs one field per pair: after certified B-spline boundaries (offsets 8–20 mm), the ten pair fields with about 6-cm cells are certified in 11–16 s each at levels of 17–26 mm, including band and regularity. The robots move to the antipodal positions on a circle of

radius 4 m under $SE(2)$ single-integrator dynamics, with start and goal angles perturbed by up to 9° and 20° to avoid the central deadlock of exactly antipodal goals. With the surface cover, all robots reach their goals after 12.5 s (Fig. 3); the minimum ground-truth distance is 0.8 mm, and the certified lower bound of the lifted barriers stays above +0.17 mm. While all five crowd the center, up to 2716 constraints are active, and the cost per step rises from 2.8 ms (median) to 22 ms (95th percentile). With the compiled fields, the robots arrive after 11.9 s at 1.2/2.5 ms per step, keep 20.0 mm, and the minimum active h is -1.1 mm, a discrete-time effect.

With the same shapes at half size, 20 random instances with poses in $[-3, 3]^2 \times \mathbb{S}^1$ compare the barriers under single-integrator dynamics (Table III). The closest-point barrier collides in 5 instances, with vertices up to 87 mm inside the other body: the closest pair of non-convex bodies jumps, the derivative is one-sided, and the constraint normal can point the wrong way just before contact. These results concern this single-closest-pair construction, the one analyzed in [10], not nonsmooth formulations over all active features. Both of our constructions and the circles never collide; the circles reach all goals, whereas the surface cover and the compiled fields deadlock in 8 and 7 instances, mostly the same ones, because exact non-convex boundaries can hook where round surrogates slide past. On the 11 instances completed by all methods, path

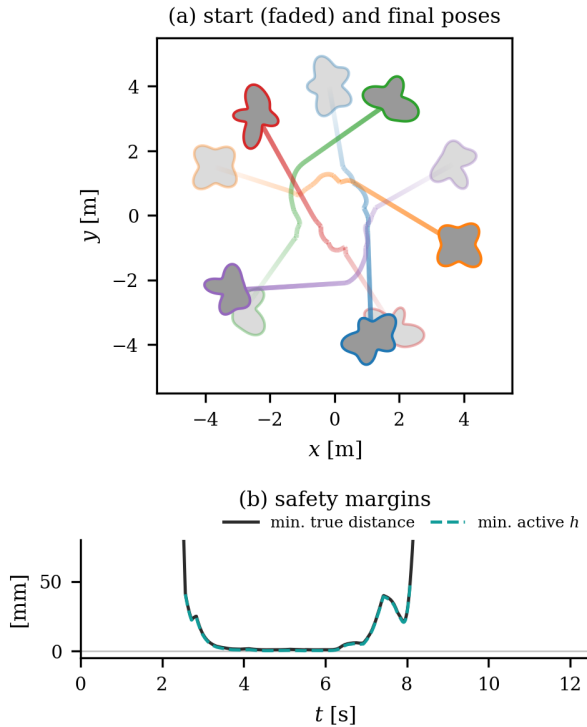


Fig. 3. Five robots with random non-convex shapes exchanging positions under the surface-cover barriers (gray: ground truth; colored: certified level curves). (a) Start (faded) and final poses with center paths, drawn from faint (start) to strong (end). (b) Minimum ground-truth distance over all pairs and certified lower bound of the lifted barriers $\tilde{h}$ over the active boxes.

TABLE III
Twenty Random Five-Robot Instances

	ours		closest	
	cover	compiled	point	circle
instances with a collision	0	0	5	0
min. true distance [mm]	+0.6	+19.7	-86.8	+42.7
min. $\tilde{h}$ [mm]	0.0	-2.4	-114.4	-
instances, all goals reached	12	13	12	20
constraints, max.	1020	10	10	10
time/step, med./max. [ms]	2.0/10.2	0.59/7.4	2.5/18.7	0.38/0.8

Horizon 15 s, step 10 ms. Negative distances: depth of the deepest vertex inside the other polygon. Cover: $\tilde{h}$ is the certified lower bound of $\tilde{h}$ over the active boxes. No step required slack. Time: median over instances of the per-instance median, and maximum; one CPU thread.

lengths differ by less than 1% and times by less than 10%, the surface cover being the slowest because its remainders restrict the speed of bodies that slide close to each other: the benefit of exact geometry appears in tight interactions such as docking. The surface cover lets the robots pass within 0.6 mm, thirty times closer than the compiled fields, and the certified lower bounds of its lifted barriers stay nonnegative; it uses up to 255 active boxes (1020 constraints), and its cost per step is about three times that of the compiled fields, with a maximum of 10.2 ms. The offline cost is 14.5 s for the five bodies, against 11–16 s for each of the ten pairs of compiled fields. In 8 instances the

compiled barrier dips below zero, by at most 2.4 mm, when several pairs are active at full speed, while the ground-truth distance stays above 19.7 mm owing to the certified levels. In unicycle swaps of two and four robots with the compiled fields, no instance collides (minimum distances 10.6 and 20.1 mm), the barrier’s input coefficient norm stays above 0.128 whenever $h < 5$ cm, and no step requires slack; head-on and diagonal swaps deadlock.

E. Static Obstacles

The compiled construction encodes a union of static obstacles in one field and one barrier: with one field for a robot among $m \in \{1, 3, 5, 7\}$ random non-convex obstacles ($122 \times 122 \times 48$ cells, 2-cm training raster, $l^* = 39\text{--}41$ mm), all runs are collision-free, and the online cost stays at 0.37–0.45 ms regardless of m , whereas the closest-point baseline needs one distance computation per nearby obstacle and grows to 2.8 ms (median) with seven.

A rocket-shaped robot (0.54 m long, nose, fuselage, two fins) escapes a curved maze (Fig. 4). The passage is our own design: three horizontal passes, each with a smooth wiggle, joined by semicircular hairpins of radius 1.1 m, with a half-width of 0.43–0.51 m, entering and leaving the 8.9×8.9 -m frame at right angles. The two wall regions are the frame minus this passage, combined through a smooth maximum of signed distances so that all corners are rounded; their ground truth consists of cubic spans, sampled into polygons with a certified chord deviation below 0.1 mm. For the surface cover, one bivariate field with 1.25-cm cells (791×791) encodes both walls, with a tightest level of 0.61 mm certified on the cubic spans (Proposition 1, with the chord bound added to the piece radius), and the rocket’s field (1-cm cells, level 1.43 mm) is covered by 383 boxes of 5 mm. The nominal controller moves toward eight waypoints whose straight connections cut into the walls by up to 10 cm, so the barriers keep the rocket off the walls: they modify the nominal input in 41% of the 3251 steps while the rocket slides along the walls to the exit in 32.5 s (path length 23.5 m), with up to 126 active boxes (504 constraints). The verified clearance over every integration interval is at least 0.5 mm, the certified lower bound of $\tilde{h}$ on the active boxes stays above 2.4 mm, and the online cost is 2.2 ms per step (median; 95th percentile 3.6 ms). A compiled field over the same maze ($160 \times 160 \times 48$ cells; rocket and walls enclosed by B-splines with 64, 5825, and 5844 control points; $l^* = 44.3$ mm certified in 45 s) also escapes, in 33.0 s, but keeps at least 17.3 mm from the walls and dips to -0.3 mm.

Sampling-based and ball-world baselines. On a fixed-orientation benchmark (a cross-shaped robot passing a cross-shaped obstacle, six approaches), adaptations of the sampling-based barrier [9] and the ball-world controller [15] do not collide either; the former matches the 29.4-mm gap of our compiled field (0.106 ms per step) only with 1024 samples (0.207 ms), and the latter needs 0.217 ms with a ten times larger input variation.

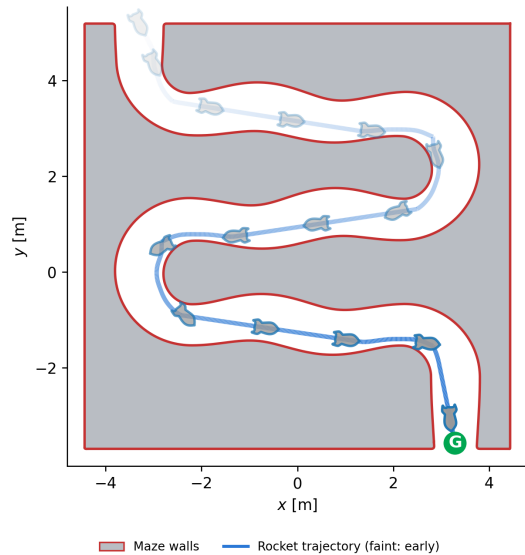


Fig. 4. Escape of a rocket-shaped robot through a curved maze with surface-cover barriers and eight goal waypoints, from the top entrance to the exit G. Gray with red outlines: walls (ground truth and certified level curve); gray with blue outlines: the rocket at intermediate times (ground truth and certified level curve); blue: path of its reference point. Poses and path are drawn from faint (start) to strong (end).

F. Franka Panda among Non-Convex Obstacles

The surface-cover barriers of Sec. V are applied to a 7-DOF Franka Panda with the link meshes and kinematics of [11]. All eight links (the hand, without fingers, and links 1–7) are constrained against three non-convex obstacles made of rounded boxes: a gate, an inverted J, and a plate carrying a 2.4-cm rod on a leg, i.e., 24 link–obstacle pairs. Each link SDF is a cubic B-spline with 12-mm cells, fitted to approximate signed distances of its mesh; its tightest level (1.8–3.8 mm) is certified on the mesh by Proposition 1 in 0.3–2 s, and its certified surface is covered by 1440–4420 boxes of side 6 mm ($r = 5.2$ mm). The obstacle fields have 2-cm cells, tightest levels of 1.9–2.9 mm, and domains extending 15 cm beyond the obstacles. The joints are velocity-controlled ($|\dot{q}_j| \leq 1$ rad/s) with a proportional joint-space nominal controller, $\gamma = 5$, a 10-ms step, and activation threshold $\eta = 3$ cm.

Start and goal configurations are drawn uniformly within 80% of the joint ranges with the hand above 0.1 m and a clearance of at least 3 cm, and we keep the first 30 pairs whose unfiltered motion collides. Safety is evaluated without sampling: the link meshes are subdivided to edges below 4 mm (same surface), and the exact obstacle SDF at each triangle centroid minus the triangle radius lower-bounds the distance between every mesh and every obstacle. As baselines, we use the 54 collision spheres of the Franka model in [11] and about 256 surface samples per link, as in the collision-avoidance experiments of [11], each with one barrier per primitive and obstacle, the exact obstacle SDF, and a 1-cm margin.

In all 30 trials, our filtered arm is collision-free (Table IV), with a certified minimum distance of 1.1 mm

TABLE IV
Franka Panda: 30 Trials Whose Unfiltered Motion Collides

	ours	spheres	samples	KKT
primitives	26,636	54	1975	24
certified guarantee	yes	no	no	no
trials with a collision	0	0	0	7
without margin	–	17	19	–
deepest penetration [mm]	–	10.2	13.4	41.9
certified min. distance [mm]	1.1	0.9	1.4	< 0
median of the minima [mm]	5.7	8.6	7.2	2.4
trials reaching the goal	17	20	23	20
trials needing slack	0	0	0	7
time/step, med./p95 [ms]	13.3/46.5	1.9/3.7	3.3/7.1	128/487
constraints, max.	34,992	21	188	5

Without the filter, all 30 trials collide. Primitives: boxes, spheres, surface samples, and closest-point pairs. Spheres and samples: exact obstacle SDF with a 1-cm margin unless stated; KKT: the certified fields and levels of ours. Distances: certified lower bounds between the link meshes and the obstacles. Time per step: median and 95th percentile over all steps of the 30 trials, one CPU thread (Python, AMD Ryzen 9 3900X).

between meshes and obstacles (median over trials 5.7 mm); no step requires slack, and the certified lower bounds of the lifted barriers stay nonnegative. The distance at contact is set by the two tightest levels (a few millimeters each), as Proposition 3 predicts; barriers at the box corners that bound ϕ_O alone kept 8.4 mm in the same trials. Within the 10-s horizon, the arm reaches the goal in 17 trials, against 23 with those corner barriers: close to an obstacle, the remainders of (7) slow the links that slide along it, and in the other trials the arm stops in front of an obstacle. The filter needs 13.3 ms per step (median; 95th percentile 46.5 ms) with up to 4374 active boxes (34,992 constraints).

With a 1-cm margin, neither baseline collides in these trials, but both come within about 1 mm of the obstacles (0.9 and 1.4 mm): the margin, not a certificate, absorbs the geometric error of the primitives. Without the margin, the spheres collide in 17 of the 30 trials, by up to 10.2 mm, because they do not enclose the meshes, and the surface samples collide in 19, by up to 13.4 mm, although every sampled barrier stays above -0.05 mm: the obstacle enters between the samples. The baselines are cheaper per step and reach the goal more often (20 and 23 trials); what the surface cover adds is that the constraint holds on the whole continuous surface, so that no margin has to be tuned. Finally, the closest-point barrier of [10], transferred to 3D with the same certified fields and levels (one barrier per link–obstacle pair at $\mathbf{x}^* = \arg \min \phi_O$ subject to $\phi_A = l_A$, SLSQP warm-started from the previous step), collides in 7 trials by up to 41.9 mm, violates its own barrier by up to 37 mm, needs slack in 7, and costs 128 ms per step: on non-convex links the closest point is only a local minimizer and jumps between surface patches, as in the planar comparison.

Threading a star-shaped tube. Fig. 5 shows a tighter static scene, arranged in an interactive tool and replayed: a tube whose hole is a 5-pointed star (tips at 24 cm, notches at 15 cm), surrounded by walls of constant 5-cm thickness and 6.7 cm deep, stands 0.65 m high in front of



Fig. 5. Franka Panda threading its forearm through a star-shaped tube (blue; 5-pointed star hole, walls of constant 5-cm thickness, 6.7 cm deep) with the surface-cover barriers of all eight links. Without the filter, the same motion cuts through the tube wall by 30.6 mm.

the arm. Its ground truth is the extruded star-minus-star region, whose exact SDF is analytic; its field has 1.5-cm cells and a tightest level of 3.9 mm. The arm starts beside the tube and a damped-least-squares IK moves the hand to the home pose, which puts the forearm through the hole. Without the filter, the motion cuts through the tube wall by up to 30.6 mm and stays in contact for 10.2 s. With it, the arm threads its forearm through the star hole and reaches the home pose after 4.1 s, with a certified clearance of at least 3.0 mm and no slack. Near the sharp edges of the star, ϕ_O varies quickly, and refining only the boxes within 1 cm lets the arm freeze at the rim; in this scene, every active box whose remainder exceeds ϑ is refined, with up to 5410 active boxes (43,280 constraints) and 96 ms per step (median; 95th percentile 109 ms).

G. Two Franka Pandas in a Shared Workspace

Two Franka Pandas face each other with bases 0.9 m apart. For each of the 64 pairs of a link of arm 1 and a link of arm 2, the certified surface of the first is bounded against the SDF of the second; both links move, so each constraint is affine in the velocities of all 14 joints (the derivative in Sec. V with the Jacobians of both arms), and one direction per pair suffices by Theorem 1. Starts and goals of both arms are drawn as before, with an initial clearance of 5 cm, and we keep the first 20 of 143 sampled pairs whose unfiltered motion brings the arms into contact (sampled mesh distance below 2 mm). The distance between the arms is bounded from below with exact point-to-triangle distances from the subdivided triangles of arm 1 to the meshes of arm 2, at the 20 steps of each trial with the smallest sampled distance. In all 20 trials the arms stay apart without slack (Fig. 6 shows one); the barriers are active ($\min h < 5$ mm) in 8 trials, the meshes come within 3 cm of each other in 16, and the certified minimum distance is 1.6 mm, the two levels of a pair of links (corner barriers on ϕ_B alone kept 10.3 mm). Both arms reach their goals in 15 trials (17 with the corner barriers). The filter needs 7.6 ms per step (median; 95th percentile 78 ms) with up to 6730 active boxes (53,840 constraints) over the 64 link pairs.

IX. Limitations

Certificates. The certificates exclude first contacts, so they protect trajectories that start collision-free, and they certify non-intersection, not a prescribed clearance (which follows by enclosing dilated ground truths). They assume exact arithmetic; our implementation uses double precision without outward rounding.

Discrete time. The guarantees hold in continuous time. With 10-ms steps, the compiled barrier dipped to -2.4 mm in random swaps, whereas the certified lower bounds of the surface cover stayed nonnegative; a sampled-data tightening is needed for the implemented controller.

Conservatism, feasibility, and cost. The safe set of the surface cover is exact up to the two certified levels, but its enforcement is not: the Taylor remainders and the Bernstein coefficients restrict the admissible inputs, so the filter stops bodies before the boundary of the safe set by an amount of second order in the box size, may slow them where the fields vary quickly, and is guaranteed feasible only outside that band. Where the third derivatives of the fields are large, as near sharp edges, finer boxes would recover speed at the price of more constraints. Their number, 2^n per active box and thousands in tight spatial scenes, sets the cost per step and its variation. The compiled field keeps one barrier and a bounded cost per pair, but its level is limited by the training targets; spatially varying levels would reduce it.

Scope. Deadlocks are not resolved, and exact geometry makes them more frequent than round surrogates. The compiled construction is limited to low-dimensional relative poses and needs one field per pair of shape types. The manipulator studies omit self-collision within an arm, the fingers, and polytopic baselines, and all experiments are in simulation.

X. Conclusion

Collision avoidance between non-convex rigid bodies reduces, through a first-contact lemma, to keeping each body's SDF above a certified level on the boundary of the other. We took this condition on the continuous level set as the safe set and enforced it through Bernstein bounds of a lifted barrier on a cover of that level set, so that

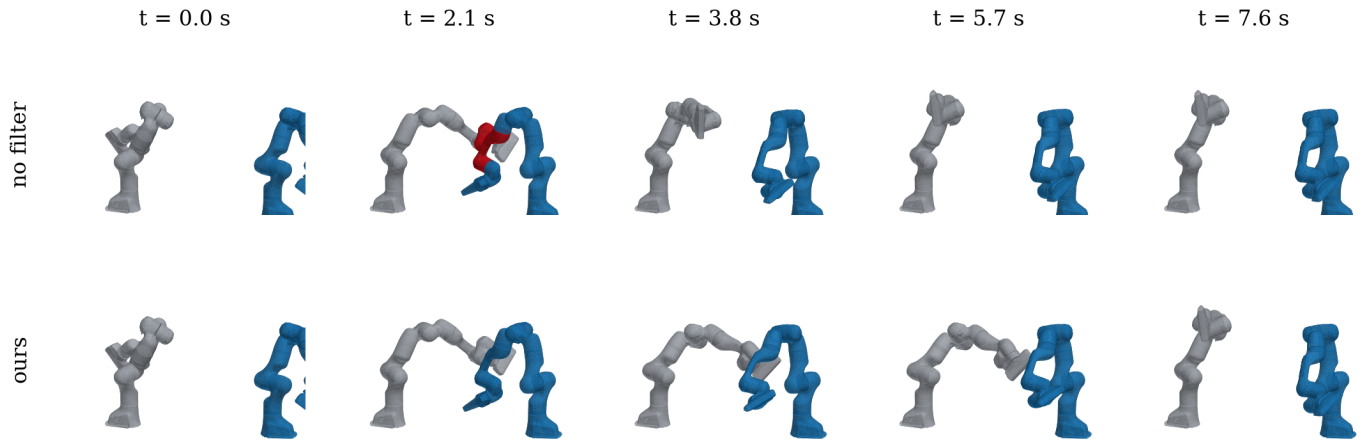


Fig. 6. Two Franka Pandas move to their goals and cross. Top: without the safety filter, the arms collide (links in contact red) and pass through each other. Bottom: with the surface-cover barriers of all 64 link pairs, arm 1 (gray) and arm 2 (blue) slide past each other and both reach their goals, 7.6 s after the start.

the finite certificate restricts only the inputs, for planar and spatial, rigid and articulated bodies; for planar pairs, we also compiled it into one configuration-space field per pair, certified by a Bernstein branch-and-bound. Neither construction computes closest points, and both certify enclosure of the true geometry. Both never collided in planar experiments, unlike a closest-point barrier, and docked where bounding circles cannot; the surface cover kept a Franka Panda off three non-convex obstacles and two Pandas apart, within the certified levels. Sampled-data guarantees, faster enforcement near contact, deadlock resolution, and hardware validation are the next steps.

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